[Working Title] Beyond Right or Wrong: Detecting and Correcting Student Misconceptions in LLM-Based Tutoring Dialogues

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Abstract

(150–250 words. Write this last, once results are in.)

Large language model (LLM)-based tutoring chatbots are increasingly used to provide scalable, personalized instruction, but most existing systems treat student responses as simply "correct" or "incorrect," offering generic corrective feedback without understanding *why* a student arrived at a wrong answer. This shallow treatment of student errors limits the pedagogical value of AI tutors, since effective human tutors diagnose the underlying misconception before intervening. In this work, we propose [SYSTEM NAME], an LLM-based tutoring chatbot that explicitly detects the misconception underlying an incorrect student response and generates feedback targeted at correcting that specific misconception, rather than the surface-level error. We evaluate [SYSTEM NAME] on [DOMAIN, e.g., introductory algebra / a specific CS topic] using [N] dialogues / [N] participants, comparing it against a generic Q&A tutoring baseline. Our results show [PLACEHOLDER: e.g., "misconception-targeted feedback led to X% higher post-test scores and Y% fewer repeated errors of the same type compared to generic feedback"]. We discuss implications for the design of pedagogically grounded AI tutors and outline directions for extending misconception detection beyond [DOMAIN].

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1. Introduction

The rapid adoption of large language models (LLMs) in education has produced a wave of AI tutoring chatbots capable of holding open-ended, natural-language dialogues with students (Gupta et al., 2025; Kumar et al., 2023). These systems promise to make one-on-one tutoring — long considered one of the most effective instructional methods but difficult to scale — available to far more learners.

However, a growing body of work has found that most LLM-based tutors, even strong ones, struggle to move beyond surface-level correctness checking. Gupta et al. (2025) found that while ChatGPT-family models could solve the large majority of college-algebra problems correctly, their tutoring dialogues frequently contained instructional errors, and only a little over half were entirely correct as *tutoring* interactions rather than just as problem-solving ones. Kumar et al. (2023) showed that LLM-generated explanations can meaningfully improve learning outcomes relative to bare correct answers — but this benefit hinges on the *quality* of the explanation, not just its presence.

Critically, an effective tutor — human or AI — does not simply mark a student's answer wrong. It asks: *why* is the student wrong? A student who answers "8" to "5 + 3 × 2" may be making an order-of-operations error; a student who says a fraction gets *smaller* when you add a positive number to its denominator may be over-generalizing "bigger denominator, smaller value" without understanding why. These are different misconceptions requiring different corrective explanations, even though both produce an "incorrect" label under a binary right/wrong framing.

Scarlatos, Baker, and Lan (2025) took a first step toward addressing this gap, showing that LLMs can be prompted to identify the knowledge component involved in each turn of a tutoring dialogue and to trace whether a student's understanding of that component is improving over time (a task they call dialogue knowledge tracing, or dialogueKT). Their method, LLMKT, significantly outperformed prior knowledge-tracing approaches at predicting whether a student's *next* response would be correct. However, their work stops at *predicting correctness* — it does not attempt to characterize *what kind of error* the student is making, nor does it close the loop by generating feedback targeted at that specific misconception.

This paper extends that line of work. We propose and evaluate [SYSTEM NAME], a tutoring chatbot that:

1. **Detects** the specific misconception underlying an incorrect student response (not just that it is incorrect),
2. **Classifies** that misconception against a domain-specific misconception taxonomy, and
3. **Generates** feedback explicitly targeted at correcting that misconception, rather than generic corrective text.

Our central hypothesis is that **misconception-targeted feedback produces better learning outcomes than generic correctness-based feedback**, even when both are generated by the same underlying LLM.

1.1 Contributions

- A misconception taxonomy for [DOMAIN], built from [data source: e.g., prior literature / a pilot study / expert tutor interviews].
- [SYSTEM NAME]: an LLM-based pipeline for misconception detection and targeted correction in tutoring dialogues.
- An empirical comparison of misconception-targeted vs. generic feedback, evaluated via [benchmark correctness testing / human-as-student study / learning gain pre-post test — pick what you'll actually run].
- A discussion of failure modes and design implications for future misconception-aware tutoring systems.

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2. Related Work

2.1 LLMs as Math/Subject Tutors

Recent work has evaluated LLMs' capability both as problem-solvers and as interactive tutors. Gupta et al. (2025) introduced a two-part evaluation methodology — benchmarking LLM problem-solving against an existing intelligent tutoring system, and having human evaluators act as students receiving tutoring — finding a gap between LLMs' raw problem-solving accuracy and the quality of their tutoring dialogues. We adapt a similar two-part evaluation approach in Section 4.

2.2 Knowledge Tracing and Misconception Analysis in Dialogue

Traditional knowledge tracing (KT) methods model a student's evolving mastery of skills over a sequence of graded responses, but have historically been applied to structured, multiple-choice-style interaction logs rather than open-ended dialogue. Scarlatos et al. (2025) extended KT into open-ended tutor-student dialogue, showing that LLMs can be used to automatically tag dialogue turns with knowledge components and correctness labels, enabling knowledge tracing directly on dialogue data. Their method (LLMKT) is the closest prior work to ours, and we treat it as our primary comparison point: where LLMKT traces *whether* a student's understanding of a skill is improving, [SYSTEM NAME] aims to characterize *what specific misunderstanding* is driving an incorrect response, and to act on that diagnosis.

2.3 Feedback Generation for Incorrect Responses

Lin et al. (2024) showed that GPT-4 can be used first to classify whether a (human tutor trainee's) response is pedagogically correct, and then to rephrase incorrect responses into improved ones that match expert-tutor quality, using few-shot prompting. While their setting is tutor-training rather than student-tutoring, their two-stage "classify, then rewrite" pipeline is structurally similar to the detect-then-correct pipeline we propose, and we adapt elements of their approach for the correction-generation stage of [SYSTEM NAME].

2.4 The Value of Explanation Quality in Learning

Kumar et al. (2023) ran large pre-registered experiments showing that LLM-generated explanations improve learning outcomes on math problems relative to seeing only the correct answer, with the largest benefit for students who attempted the problem before seeing the explanation. Notably, they found that even LLM explanations containing arithmetic errors still produced learning gains better than no explanation at all — suggesting that the *framing and structure* of an explanation may matter as much as its raw correctness. This motivates our focus on *targeted* rather than merely *correct* feedback.

2.5 Gap Addressed by This Work

Taken together, prior work establishes that (a) LLM tutoring dialogues often fall short of expert quality despite strong raw problem-solving ability, (b) LLMs can be used to trace whether a student's skill mastery is improving over a dialogue, and (c) LLMs can be prompted to rewrite an identified incorrect response into a better one. No prior work we are aware of

combines these into a single pipeline that diagnoses the *specific misconception* behind an error in real time and generates feedback explicitly targeted at that misconception, then evaluates whether this produces better learning outcomes than generic feedback. This is the gap [SYSTEM NAME] addresses.

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3. System Design: [SYSTEM NAME]

3.1 Overview

[SYSTEM NAME] consists of three stages operating on each incorrect student turn in a tutoring dialogue:

1. **Error Detection** — determine whether the student's response is correct or incorrect (and, optionally, partially correct).
2. **Misconception Classification** — given an incorrect response, classify it against a predefined taxonomy of common misconceptions for the relevant knowledge component, or flag it as "novel/unclassified" if it doesn't match any known category.
3. **Targeted Correction Generation** — generate a tutoring response that directly addresses the identified misconception (e.g., naming the faulty rule the student seems to be applying, and walking through a counter-example), rather than a generic "that's incorrect, try again" or a generic worked solution.

(Insert a system architecture diagram here — Stage 1 → Stage 2 → Stage 3 → Student, with a knowledge-component/misconception log feeding back into a running student model.)

3.2 Misconception Taxonomy

For the domain of [DOMAIN], we construct a taxonomy of common misconceptions from [SOURCE: e.g., prior education-research literature on error patterns in this domain / analysis of a pilot dataset of student errors / structured interviews with subject-matter tutors]. Example taxonomy entries:

Misconception ID	Description	Example incorrect response	Underlying faulty rule
M1	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
M2	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
M3	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]

(Fill this in with the actual misconceptions relevant to your chosen subject domain — this table is the backbone of your whole system and paper.)

3.3 Error Detection (Stage 1)

We use [zero-shot / few-shot] prompting of [MODEL, e.g., GPT-4o] to classify each student response as correct, partially correct, or incorrect, following the classify-then-act approach used by Lin et al. (2024) for trainee-response classification. [Describe your prompt design, and whether you validate this stage against human-annotated ground truth, as in Scarlatos et al. (2025) and Henkel et al. (2025).]

3.4 Misconception Classification (Stage 2)

Given an incorrect response, we prompt the model with the misconception taxonomy (Section 3.2) and ask it to select the best-matching misconception category, or return "unclassified." [Describe: in-context taxonomy injection vs. fine-tuning; how you handle ambiguous or multi-misconception cases; how classification accuracy is validated against human expert labels.]

3.5 Targeted Correction Generation (Stage 3)

Once a misconception is identified, we generate feedback using a prompt that explicitly (a) names the underlying faulty rule/assumption without being condescending, (b) provides a counter-example or contrasting case that breaks that specific rule, and (c) invites the student to re-attempt the original problem. This is designed to go beyond generic "here's the correct answer" feedback, consistent with evidence that explanation *structure*, not just correctness, drives learning gains (Kumar et al., 2023).

3.6 Dialogue and Student Model

[Describe how misconception history is tracked across a session — e.g., a running log per student per knowledge component, similar in spirit to the dialogueKT approach of Scarlatos et al. (2025), but tracking misconception *type* rather than only correctness.]

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4. Evaluation Methodology

Following the two-part evaluation approach of Gupta et al. (2025), we assess [SYSTEM NAME] along two axes: (a) the accuracy of the underlying detection/classification pipeline, and (b) the quality and learning impact of the tutoring interaction as a whole.

4.1 Dataset

[Describe your data: e.g., number of dialogues/problems, domain, how student responses were collected — real classroom logs, a controlled study, or synthetic student personas.]

- Domain: [PLACEHOLDER]
- Number of dialogues / student responses: [PLACEHOLDER]
- Number of participants (if human study): [PLACEHOLDER]
- Ground-truth misconception labels obtained via: [PLACEHOLDER — e.g., expert annotation, following an inter-rater reliability process]

4.2 Baseline

We compare [SYSTEM NAME] against a **generic Q&A tutoring baseline**: an LLM tutor prompted only to determine correctness and provide a standard corrective explanation, without misconception-specific targeting. This isolates the effect of misconception-targeted feedback from the general capability of the underlying LLM.

4.3 Metrics

Pipeline accuracy (Stages 1–2):

- Error detection accuracy (correct/incorrect), compared against human-expert labels, reported via F1 and/or Cohen's κ (as in Henkel et al., 2025; Lin et al., 2024).
- Misconception classification accuracy against expert-labeled ground truth.

Tutoring quality and learning impact (Stage 3 + full system):

- Human-as-student qualitative evaluation of feedback quality, following the coding approach of Gupta et al. (2025).
- Pre/post-test learning gains, comparing students who receive misconception-targeted feedback vs. generic feedback ([WITHIN-SUBJECT / BETWEEN-SUBJECT] design).
- Rate of repeated errors of the same misconception type within a session (a proxy for whether the feedback actually resolved the underlying confusion, rather than just the surface error).

4.4 Procedure

[Describe study procedure: recruitment, session structure, randomization to condition, any IRB/ethics approval process if applicable.]

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5. Results

(To be completed once data collection is done. Suggested structure below — replace with real numbers/tables/figures.)

5.1 Pipeline Accuracy

[PLACEHOLDER: Table of error-detection and misconception-classification accuracy, F1, κ.]

5.2 Tutoring Quality

[PLACEHOLDER: Human evaluation results — e.g., % of dialogues rated as high-quality, breakdown by misconception type.]

5.3 Learning Outcomes

[PLACEHOLDER: Pre/post-test comparison between misconception-targeted and generic-feedback conditions, with effect sizes.]

5.4 Qualitative Findings

[PLACEHOLDER: Illustrative dialogue examples showing successful and unsuccessful misconception correction.]

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6. Discussion

[Once results are in, discuss: Did misconception-targeted feedback outperform generic feedback? Under what conditions did it fail (e.g., unclassified/novel misconceptions, ambiguous student language)? How does pipeline accuracy propagate into final learning outcomes — i.e., does a wrong misconception classification lead to actively harmful feedback, and how often does this happen?]

6.1 Comparison to Prior Work

Relate your findings back to Scarlatos et al. (2025): did going beyond correctness-tracing to misconception-specific diagnosis meaningfully change learning outcomes, or mainly change perceived feedback quality? Relate to Kumar et al. (2023)'s finding that even flawed explanations can help — does this still hold when the *misconception classification itself* is wrong?

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7. Limitations

- Misconception taxonomy is limited to [DOMAIN] and may not generalize to other subjects without re-derivation.
- [Sample size / demographic] limitations of the study.
- Reliance on [MODEL NAME]'s underlying capabilities; results may not transfer to smaller or open-weight models.
- Ground-truth misconception labels are themselves subject to annotator disagreement (report your inter-rater reliability here).

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8. Conclusion and Future Work

[Summarize contribution and headline result in 3–4 sentences.] Future work includes extending the misconception taxonomy to additional domains, exploring automatic (rather than manually curated) misconception taxonomy induction from large-scale student response logs, and studying long-term retention effects of misconception-targeted tutoring beyond a single session.

References

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(Add any additional citations you draw on — dblp.org and the AIED 2025 proceedings page have full BibTeX entries for all AIED papers if you need exact formatting.)

Notes for you (remove before submission)

- Every `[PLACEHOLDER]` and bracketed instruction above needs your actual project details — I've built the skeleton and argument structure, but the taxonomy, dataset, and results are yours to fill in since I don't have access to your system or data.
- The paper is structured to directly position your work as extending Scarlatos et al. (2025) and using Lin et al. (2024) and Gupta et al. (2025) as methodological templates, per what we discussed.
- Target venue: given your timeline, AIED 2026 (Seoul, June 29–July 3) or a workshop like BEA are reasonable targets — check their specific formatting/page-limit requirements (LNCS format for AIED) before final submission.